

JULIAN KALLIGONIS

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U.S. Citizen - Eligible for Security Clearances

EDUCATION

The Pennsylvania State University

Penn State College of Engineering

Bachelor of Science in Mechanical Engineering

University Park, PA

May 2028

GPA: 3.4/4.0

SKILLS / HONORS & CERTIFICATIONS

Skills: SolidWorks, Fusion 360, FEA, MATLAB, OpenRocket, Betaflight, AutoCAD, Bluebeam, Structural Analysis, Composites/carbon fiber layup, 3D printing, Soldering, Excel

Honors & Certifications: (CSWA) Certified SolidWorks Associate, 2nd Place & People's Choice Award-Penn State ASME Battle Bots, Dean's list (PSU College of Engineering)

INTERNSHIPS / ENGINEERING EXPERIENCE

American Society of Mechanical Engineers (ASME)

Design Team Project Lead (VTOL Aircraft)

State College, PA

May 2026 – Present

- Led a team of engineering students to research, design, manufacture, and automate a VTOL aircraft
- Managing a \$1,500 project budget and coordinating subteams spanning airframe, propulsion, avionics, and payload systems
- Defined project scope, created a timeline, and conducted a personal drone project in preparation for the start in September

Design Team Engineer (Battle Bots - Weapon system)

August 2025 – April 2026

- Reduced internal stress by over 50% by analyzing several FEA simulations of the blade and redesigning
- Presented FEA-driven design and stress analysis to Lockheed Martin and TE Connectivity engineers at project showcase
- Created 14 different design iterations in CAD for a weapon blade made of AR500 steel

Nittany Rocket Labs

Certification Team

State College, PA

January 2026 – March 2026

- Designed a 36 in, 3 in diameter H-class rocket in OpenRocket, performing stability (CG/CP) and performance simulations
- Modeled and 3D printed a custom PLA nose cone; generated DXF files from CAD to laser cut precision wooden fins
- Assembled and integrated full flight vehicle for launch to obtain Level 1 High Power Rocketry Certification

IREC Structures Team

March 2026 – Present

- Building carbon fiber airframe components for a student-designed rocket competing at the 2026 Spaceport America Cup (IREC) with a 30,000 ft target altitude
- Contributed to much of the composite manufacturing with hand layups using 3K and 6K carbon fiber to fabricate the motor tube, fins, and body tube

Paradigm Consulting Engineers

Mechanical Engineering Intern

West Chester, PA

May 2026 – August 2026

- Produced complete HVAC construction documents in AutoCAD for over 30 commercial, residential, and educational projects (15,000–30,000 SF); includes duct layouts, VAV/FPB schedules, and sizing verified against static pressure targets
- Performed CFM balancing for VAV/FPB arrays, ventilation load calculations per ASHRAE 62.1, and electrical panel load scheduling across multiple concurrent projects
- Collaborated directly with licensed engineers in project meetings and contributed to MEP drawing set reviews in Bluebeam

5" FPV Drone

Personal Project

West Chester, PA

July 2026 – August 2026

- Independently designed, built, and tuned a 5" 6S FPV quadcopter, integrating flight controller, ESCs, and brushless motors through hand-soldered power connections and firmware configuration in Betaflight software
- Diagnosed and resolved electrical integration issues (wire gauge selection, connector sizing, and VTX overheating) to ensure reliable high-current power delivery under flight loads

CAMPUS INVOLVEMENT

Penn State Powerlifting Team

State College, PA

- Committed to rigorous training schedule (5x/week) while fulfilling yearly community service requirements